

PRESS RECORD

COMPX341 - Advanced Software Engineering Design



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Before we start ...

- Guest speaker this Friday
- Please come along in person



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DevOps – CI/CD Pipelines

COMPX₃₄₁

Overview

- What is CI/CD?
- What are CI/CD pipelines?
- CI/CD pipeline example

What is CI/CD

What is CI/CD?

“CI/CD is a method to frequently deliver apps to customers by introducing automation into the stages of app development. ”

<https://www.redhat.com/en/topics/devops/what-is-ci-cd>

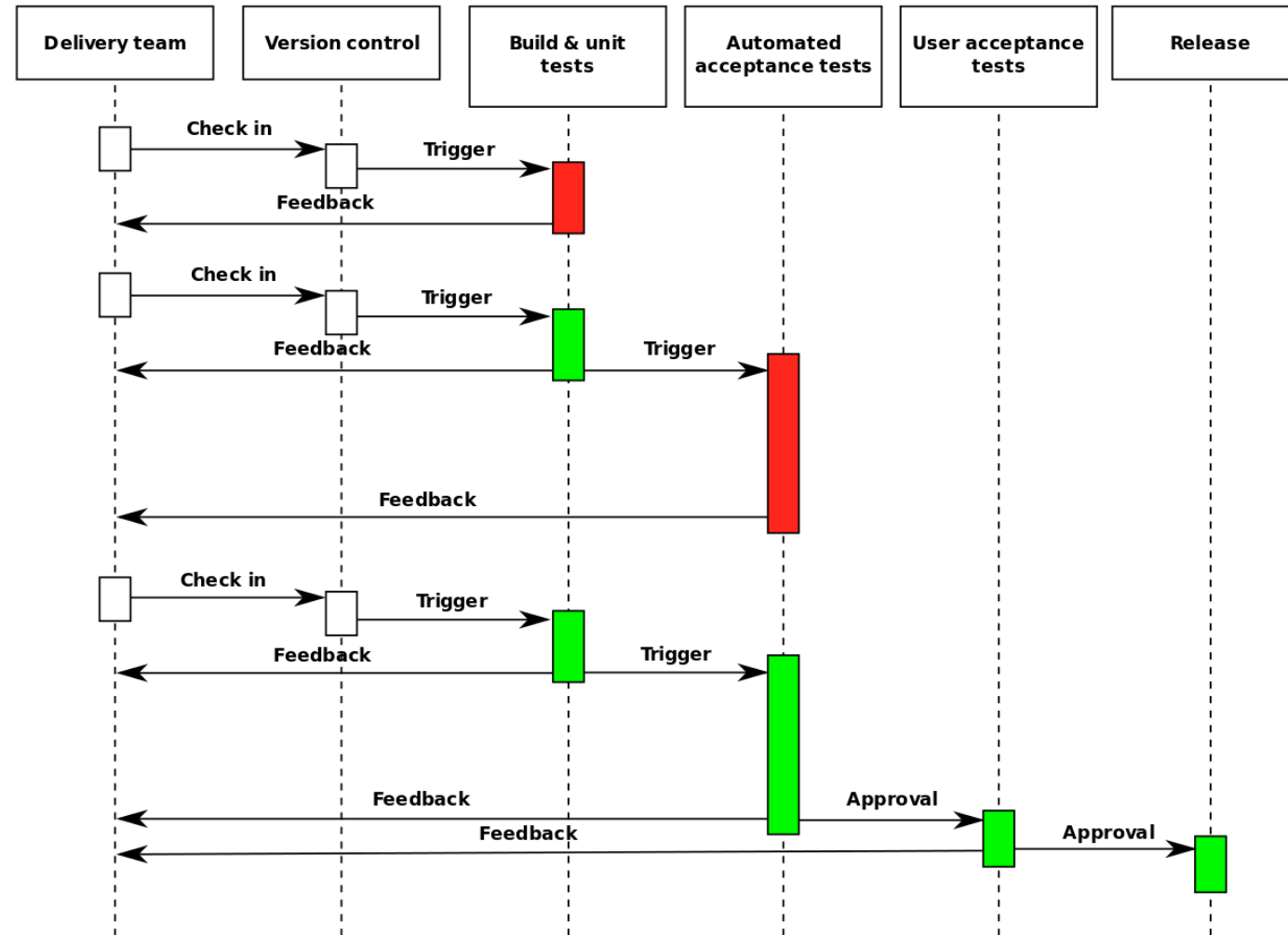
CI/CD stands for Continuous Integration and Continuous Delivery/Continuous Deployment.

What is CI/CD?

Continuous Integration / Continuous Delivery / Continuous Deployment

- Continuous integration
 - Frequent merging of separate dev/team code several times a day
- Continuous delivery/Continuous Deployment
 - Aim for constant system availability
 - Aims to increase the rate of producing new versions
 - Aims to shorten the dev/testing time
 - Automates processes

CI/CD example



CI/CD Pipelines

CI/CD Pipelines

- A CI/CD pipeline is a series of steps that are taken/performed in order to deliver software
- CI/CD pipelines focus on improving delivery using a DevOps or SRE approach
- Note, it is possible to manually execute each step in a CI/CD pipeline. However, the true value of CI/CD pipelines is their automation!

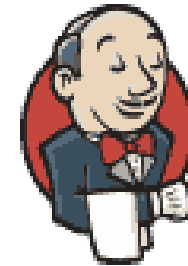
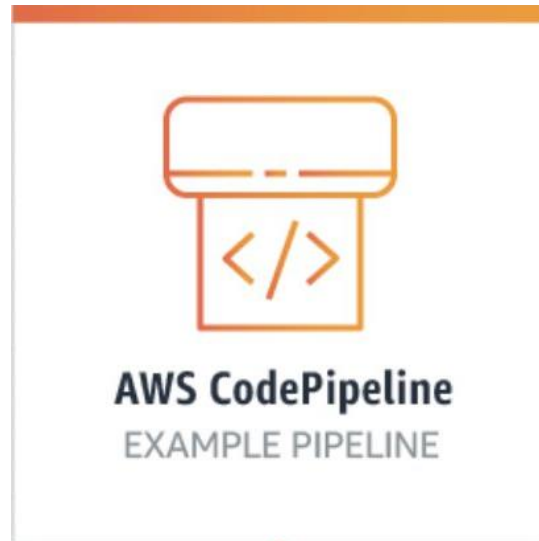
CI/CD Pipelines

- Typical CI/CD pipeline stages include:
 - Build (compiling the application)
 - Test (code testing, i.e. unit tests)
 - Release (committing to a repo)
 - Deploy (deployed to production)

CI/CD Pipelines



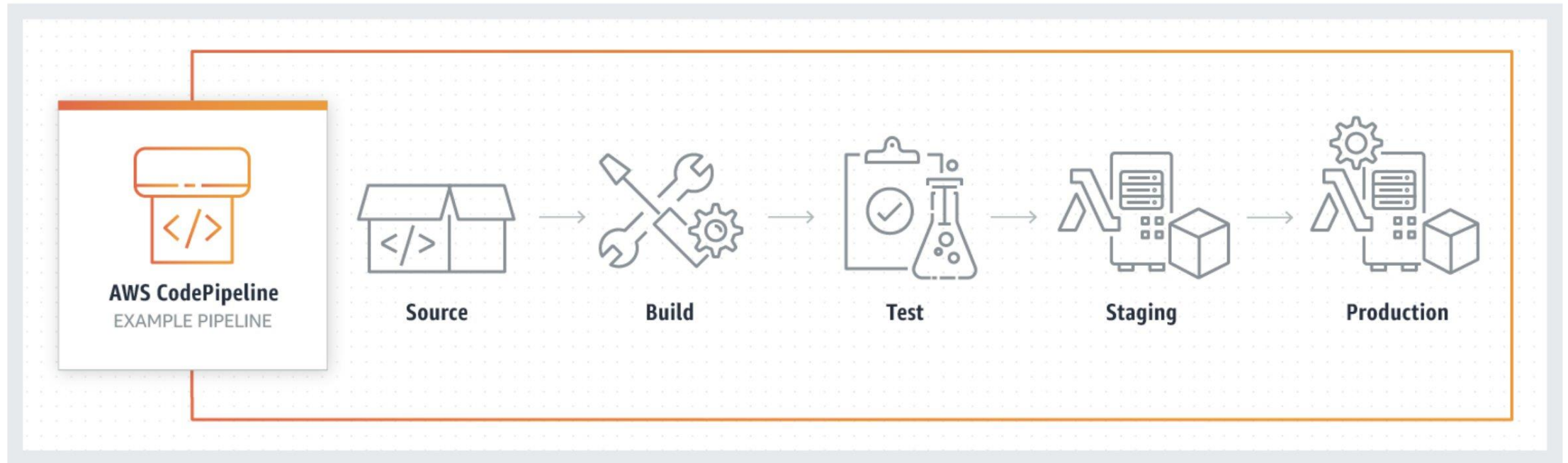
CI/CD Pipelines



Jenkins

CI/CD Pipeline Example

CI/CD pipelines in AWS



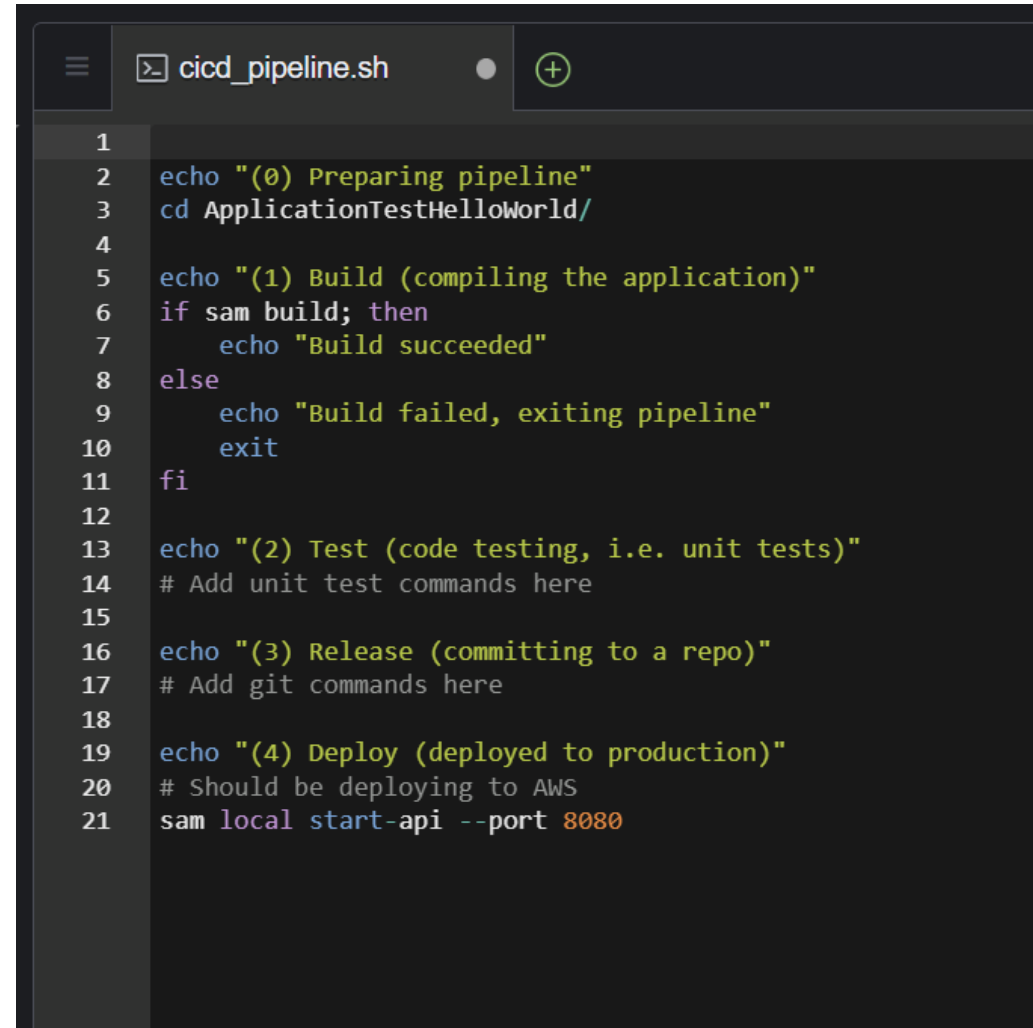
CI/CD pipelines for the command line (Linux)

You can also write a bash script

What is a bash script?

- A plain text file that contains a set of CLI commands
- The file extension is .sh
- For example ...

CI/CD pipelines for the command line (Linux)



```
1
2 echo "(0) Preparing pipeline"
3 cd ApplicationTestHelloWorld/
4
5 echo "(1) Build (compiling the application)"
6 if sam build; then
7     echo "Build succeeded"
8 else
9     echo "Build failed, exiting pipeline"
10    exit
11 fi
12
13 echo "(2) Test (code testing, i.e. unit tests)"
14 # Add unit test commands here
15
16 echo "(3) Release (committing to a repo)"
17 # Add git commands here
18
19 echo "(4) Deploy (deployed to production)"
20 # Should be deploying to AWS
21 sam local start-api --port 8080
```

CI/CD pipelines for the command line (Windows)

You can also write a batch script (Windows)

What is a batch script?

- A plain text file that contains a set of CLI commands
- The file extension is .bat
- For example ...

CI/CD pipelines for the command line (Windows)

```
run_empty.bat x
1  @echo off
2
3  echo:
4  echo (0) Preparing pipeline
5
6  echo:
7  echo (1) Build (compiling the application)
8  javac -cp "junit-platform-console-standalone-1.8.2.jar" *.java
9  IF %ERRORLEVEL% NEQ 0 (
10     echo Build failed, exiting pipeline
11     set /p DUMMY=Hit ENTER to finish...
12     EXIT /B
13 ) ELSE (
14     echo Build successful
15 )
16
17 echo:
18 echo (2) Test (running the test suite)
19 rem Add unit test commands here
20
21 echo:
22 echo (3) Release (committing to repo)
23 rem Add git commands here
24
25 echo:
26 echo (4) Deploy (deployed to production)
27 rem Add run commands here
28
29 echo:
30 set /p DUMMY=Hit ENTER to finish...
```

CI/CD pipelines for the command line (Linux/Windows)

You can also write a bash/batch script

What pipeline should (at minimum) include

1. Some form of preparation
2. Build (compiling the software)
3. Test (e.g. running regression tests)
4. Release (committing to a repository)
5. Deploy (optional, depends on the pipeline)

CI/CD pipelines for the command line (Windows)

- Okay, let's give it a try ...

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